

DOMAIN CONTRASTIVE LEARNING FOR MULTI-SITE ALZHEIMER'S DISEASE CLASSIFICATION

LINDA DELALI FIASAM^{1*}, YUNBO RAO¹, COLLINS SEY¹, ESTHER STACY E. B. AGGREY¹, SETH LARWEH KODJIKU², JUDITH AYEKAI BROWNE³, JULIANA MANTEBEA DANSO¹, CHIAGOZIEM C. UKWUOMA⁴, EMMANUEL S. A. GYARTENG³

¹School of Information and Software Engineering, University of Electronic Science and Technology of China

²School of Computer Science and Technology, Zhejiang Gongshang University

³School of Computer Science and Technology, University of Electronic Science and Technology of China University

⁴College of Nuclear Technology and Automation Engineering, Chengdu University of Technology

E-MAIL: linda.fiasam@std.uestc.edu.cn

Abstract:

In the field of medical imaging, accurately classifying Alzheimer's Disease (AD) poses a substantial challenge, primarily due to the inherent variability in imaging data across different clinical sites. This paper introduces a novel approach, Domain Contrastive Learning (DCL), to address this challenge by enhancing the robustness and transferability of AD classification models using multi-site Magnetic Resonance Imaging (MRI) data. DCL leverages a unique contrastive learning technique within an autoencoder's latent space, enabling the extraction of domain-invariant features essential for accurate diagnosis. We rigorously evaluate the proposed network on brain images from three distinct imaging sites, employing a Domain Generalization (DG) training protocol to ensure robustness and generalizability across varying data sources. Our model is evaluated on three binary classification tasks: AD vs. Normal Control (NC), AD vs. Mild Cognitive Impairment (MCI), and MCI vs. NC. Results demonstrate that DCL outperforms existing methods, especially in AD vs. NC classification, achieving higher accuracy, sensitivity, and specificity with 0.51%, 1.54% and 1.68% increments respectively. This underscores DCL's potential as a robust and generalizable method for enhanced AD diagnosis, marking a significant step forward in medical imaging challenges in a multi-site context.

Keywords:

Domain generalization; contrastive learning; Alzheimer's classification; domain shift; deep learning

1. Introduction

Alzheimer's disease (AD), a prevalent cause of dementia, is marked by a gradual deterioration of cognitive and behavioral functions, leading to an inexorable reduction in the capacity to engage in everyday tasks necessary for daily life. The increasing prevalence of this condition in the

United States from 5.7 million in 2018 to projected figures for 2040 emphasizes the need for innovative diagnostic methods [1]. Neuroimaging techniques, such as Magnetic Resonance Imaging (MRI), have become essential for identifying the subtle changes in the brain that indicate the progression of AD. This is due to its non-invasive nature and ability to provide detailed visual representations of cerebral structures [2]. However, the existence of variability in multi-site MR images poses a significant challenge to achieving consistency in diagnostic evaluations. This heterogeneity arises from variations in acquisition protocols, scanners and the subjective nature of manually designating brain regions of interest (ROI). To alleviate these issues, the automation of AD detection using advanced machine learning algorithms has attracted significant attention in recent years.

Conventional machine learning methods, like Support Vector Machines (SVM), have traditionally served as feature extractors, delineating characteristics from MRIs to construct diagnostic models [3]. Nevertheless, these techniques are often criticized for their inefficiency and reliance on expertise for feature extraction, resulting in suboptimal AD classification accuracy. Conversely, deep learning methods have revolutionized the area of medical imaging by autonomously extracting intricate features from extensive datasets, enhancing classification precision without the prerequisite of manual feature engineering. Gupta et al.[4] utilized sparse autoencoders and convolutional neural networks (CNNs) for categorizing MRI scans, while Liu et al.[5] leveraged multitask CNNs for concurrent hippocampal segmentation and AD classification. Further, Lian et al. [6] introduced a hierarchical fully convolutional network (H-FCN) to integrate atrophy localization with feature extraction for AD diagnosis. Additionally, the incorporation of attention mechanisms

within deep learning frameworks has accentuated the ability to selectively focus on salient features, greatly enhancing AD predictive models[7], [8].

The advent of multi-center MRI data collection has paved the way for diagnostic models that exhibit greater generalizability across diverse clinical settings. Yet, the variability among multi-site data remains a significant challenge, exacerbated by differences in imaging equipment, settings, and patient-related factors across institutions. While conventional preprocessing methods try to reduce these variations, there still exists an element of subjectivity that could hinder the accuracy of AD prediction models when applied to new datasets. This problem is referred to as domain shift.

Domain generalization (DG) [9], an emerging field in medical imaging, has been explored to build robust models that can handle variations across distinct data domains, such as different acquisition protocols or MRI scanners utilized in multi-site studies. The main goal of domain generalization techniques is to mitigate the performance differences that occur when a model, which has been trained on data from a particular source, is applied to unobserved data from various distributions[10]. This is important in the field of medical imaging, where variations in imaging equipment, patient demographics, or image acquisition protocols may cause models trained on data from one site to give suboptimal performance when deployed on a previously unseen data during training. Several DG methods for medical image analysis have been proposed. However, a few have explored its application in multi-site AD prediction [11].

In this work, we propose Domain Contrastive Learning (DCL) for multi-site Alzheimer's disease classification, a method designed to overcome domain-specific variabilities in multi-site MR images. DCL effectively learns domain-invariant features within the latent space using a binary pretext task, aligning feature distributions from various sites to enhance the model's generalization performance. Our approach is evaluated across MRI datasets from multiple imaging sites, demonstrating its generalization capabilities.

2. Method

2.1. Domain Contrastive Learning

We define the problem of this work in terms of DG where the objective is to learn a robust model from source domains $D = \{D_1, D_2, \dots, D_S\}$ each representing MR images from different sites as shown in Fig.1 and generalize to an unseen target domain D_T . For each domain D_S , X_S^i represents the i -th MRI scan and Y_S^i is its corresponding label labeled as NC, MCI and AD. We aim to learn a function

$f: \mathbb{R}^D \rightarrow \mathbb{C}$ parameterized by θ , predicting the class label for any given MRI scan, regardless of domain shifts. To ensure model generalization across various MRI scans, we focus on minimizing the empirical classification loss across the source domains. This involves a regularization strategy to encourage domain-invariant feature learning. The loss function is formulated as:

$$\min_{\theta} \sum_{s=1}^S \mathbb{E}_{(X_S^i, Y_S^i) \sim D_S} [\mathcal{L}(f_{\theta}(X_S^i), Y_S^i)] + \lambda \mathcal{R}(\theta) \quad (1)$$

where $\mathcal{L}$ denotes the classification loss, $\mathcal{R}$ represents a regularization term that mitigates domain-specific feature learning, and λ is a hyperparameter balancing the two terms.

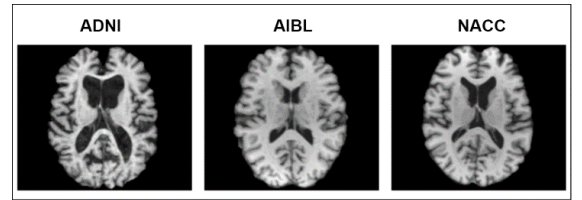


Fig. 1 A sample of MR images from different sites

We introduce a Domain Contrastive Learning (DCL) method based on self-supervised contrastive learning. This method focuses on three binary classification tasks: AD vs. NC, AD vs. MCI, and MCI vs. NC, using multi-site MR images. The DCL approach learns domain-invariant representations from these images. The overall architecture of our proposed method is shown in Figure 2.

Let E_{ϕ} be an encoder that maps input samples $\{x_i^d, x_j^d\}$ from the same or different domains belonging to two different classes (e.g., AD vs NC) in the source domain training set into a latent space Z . In this space, each instance x_i^d is represented as a latent feature vector z_i . To ensure these features are domain-invariant, we apply self-supervised contrastive learning. Specifically, we define the positive pair as latent features belonging to the same class and not necessarily the same domain denoted as $\{z_i, z_i^+\}$. This encourages the model to learn features that are invariant to the domain but specific to the class. For instance, an AD sample from one MRI site is paired with another AD sample from a different site. The negative pair are defined as features belonging to different classes (e.g., AD vs. NC or MCI). The negative pairs can be from the same or different domains denoted as $\{z_i, z_j^-\}$. The contrastive loss is calculated using the cosine similarity metric. For each positive pair $\{z_i, z_i^+\}$, the model aims to maximize the similarity. Conversely, for each negative pair $\{z_i, z_j^-\}$, the model seeks to minimize the similarity. Thus, following the standard formula of self-supervised contrastive loss[12], we define our DCL loss as:

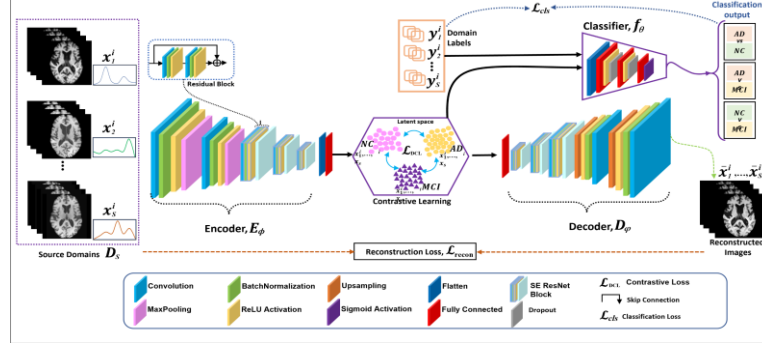


Fig.1 Over all architecture of the proposed DCL. DCL uses contrastive learning in the latent space to learn domain-invariant features based on a binary pretext task. These features are then sent to the classifier for AD classification and the decoder for reconstruction of the images

$$\mathcal{L}_{DCL} = - \sum_{i=1}^{2N} \log \frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\sum_{j=1}^{2N} \mathbb{I}_{[j \neq i]} \exp(\text{sim}(z_i, z_j^-)/\tau)} \quad (2)$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity, z_i and z_i^+ are the positive pair of examples, τ is a temperature scaling parameter, and $\mathbb{I}_{[j \neq i]} \in \{0, 1\}$ is an indicator function. The domain-invariant latent representations are simultaneously sent as inputs to a decoder D_ϕ for reconstruction, $\hat{x}_i^d$ and a classifier, f_θ for the three binary classification tasks. The reconstruction loss used to reconstruct the images is given as:

$$\mathcal{L}_{recon} = \|x_i^d - \hat{x}_i^d\|_2 \quad (3)$$

domain labels $y \in Y_s^i$ are also sent as inputs to the classifier. Thus, to train the robust classifier, f_θ we ask it to classify domain i instances from the latent space irrespective of the site differences. Hence, the following DG classification loss is minimized:

$$\mathcal{L}_{DG} = \min_{\theta, \phi} \mathbb{E}_{D_i \sim D} [\mathbb{E}_{(x_i, y_i) \sim D_i} [\mathcal{L}_{cls}(y_i, \phi(\theta(x_i)))] + \lambda_1 \mathcal{L}_{DCL} + \lambda_2 \sum_k |\theta_k|] \quad (4)$$

where $\mathcal{L}_{cls}$ is the binary cross-entropy loss for the classification task. λ_1 and λ_2 are hyperparameters that balance the classification loss and the L1 regularization term, $\sum_k |\theta_k|$, which is the sum of the absolute values of the model parameters θ . This form of regularization promotes sparsity in the model parameters, potentially leading to a more interpretable model that can better generalize to new data. Finally, the total loss to train our model is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{DG} + \lambda_3 \mathcal{L}_{recon} \quad (5)$$

Where λ_3 is a hyperparameter that regularizes the reconstruction loss $\mathcal{L}_{recon}$. Also, $\mathcal{L}_{total}$ is the total loss.

2.3. Network Architectures

Our proposed DCL framework features an encoder-decoder structure augmented with a classifier, specifically designed for multisite Alzheimer's Disease classification. The encoder utilizes 3D convolutional layers arranged in two "Convolution-Batch Normalization-ReLU" blocks for feature extraction, followed by max pooling to streamline the spatial dimensions and optimize computation. Three "Squeeze-and-Excitation" (SE) ResNet [13] blocks refine the feature maps through channel-wise recalibration, leading to a compressed yet informative latent space. This space is where our model employs contrastive loss to differentiate similar from dissimilar image representations, enhancing the discriminatory ability. The decoder reverses the encoder's structure with upsampling and 3D convolutions to reconstruct the original MRI scans. Branching off from the latent space, the classification network with three fully connected layers employs, dropout function, ReLU and sigmoid activations. It is optimized using binary cross-entropy loss to distinguish between different diagnostic categories. This streamlined approach ensures our model's efficacy in AD classification across varied imaging datasets.

3. Experiments and Results

3.1. Dataset and preprocessing

We evaluate the proposed network on brain images from three imaging sites: Alzheimer's Disease Neuroimaging Initiative (ADNI)¹, The National Alzheimer's Coordinating Center (NACC)² and The Australian Imaging, Biomarkers and Lifestyle (AIBL)³. The ADNI dataset constitutes a total of 600 scans. NACC encompasses 730 scans. Each center provided structural T1-weighted and T2-

Table 1. Characteristics and composition of the multicenter MRI scans. The values are depicted as averages. MMSE = Mini Mental State Examination ; CDR = Clinical Dementia

Characteristics	ADNI				AIBL				NACC			
	AD	MCI	NC	P-value	AD	MCI	NC	P-value	AD	MCI	NC	P-value
	(n=200)	(n=250)	(n=150)		(n=220)	(n=190)	(n=150)		(n=250)	(n=230)	(n=150)	
Sex(M/F)	139/69	140/110	90/60	0.32	140/80	120/70	90/60	0.32	150/100	150/80	190/60	0.32
Age(y)	74.5±6.0	75.7±7.0	71.4±6.2	0.30	75.2±7.2	75.0±7.3	74.4±6.0	0.30	74.6±10.3	75.7±9.3	72.7±11.2	0.30
CDR	1.2±0.6	0.5±0.2	0±0	< 0.02	1.2±0.6	0.5±0.2	0±0	< 0.02	1.2±0.6	0.5±0.2	0±0	< 0.02
MMSE	18.09±0.05	27.04±1.7	29.10±1.0	< 0.02	24.09±2.15	26.09±0.05	28.04±1.18	< 0.02	24.09±2.15	26.09±0.05	28.04±1.18	< 0.02

weighted MRI scans. These images were standardized to a uniform resolution of 1mm³. Intensity normalization brought image values to a range between 0 and 1. The dataset's characteristics are summarized in Table 1. All scans were resized to 128 × 128 × 64. The samples of each domain were randomly divided into 80% training data and 20% testing data. Data augmentation is used to increase the dataset and improve the model's generalization capabilities by introducing variations. Techniques used include random rotation of images between an angle range of -45 to 45, random scaling of images to simulate zoom in-out effects and Gaussian noise injections.

3.2. Experimental setup and implementation details

We adopt the DG training protocol [14] of leave-one-domain-out cross-validation, i.e., training on two source domains and testing on the remaining unseen domain. SGD optimizer was used to optimize the network loss during training and the accompanied hyper-parameters were set as follows: learning rate, $\alpha = 1.0 \times e^{-4}$, a weight decay of $5e^{-4}$ and a momentum of 0.9. We also train the model for 300 epochs with batch size of 2. For the weights in the total loss function, we set $\lambda_1, \lambda_2, \lambda_3$ to 1.0, 0.001 and 0.1 respectively to balance the scales of the loss terms. These values were empirically chosen after some experimentation. We compare our proposed method's performance with four other methods. All the experiments are executed in TensorFlow framework on NVIDIA RTX 3060 GPU with 12GB RAM. The performance of our proposed model was evaluated using three performance metrics: Accuracy (Acc), Sensitivity (Sen), and Specificity (Spe). These metrics are defined below:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$Sen = \frac{TP}{TP + FN} \quad (7)$$

$$Spe = \frac{TN}{TN + FP} \quad (8)$$

where TP, TN, FP, and FN represent the True Positive, True Negative, False Positive, and False Negative values respectively.

3.3. Performance Evaluation

Our performance evaluation centers on the method's ability to generalize across different sites, a crucial aspect in Alzheimer's classification. We compared it with four reference methods (Luo et al.[15], Gupta et al.[4], Bron et al.[11] and Cobbinah et al.[16]) for multi-site Alzheimer's classification. Our comparative analysis focused on three distinct binary classification tasks: AD vs. NC, AD vs. MCI, and MCI vs. NC. Table 2 shows the results obtained for all three tasks across different target sites which highlights the generalization capability of our proposed method when compared to the reference methods. This superior performance is particularly evident in the AD vs. NC classification task on all sites. Fig. 3 and 4 show the training accuracy and loss performance of DCL on source domains across the three binary classification tasks.

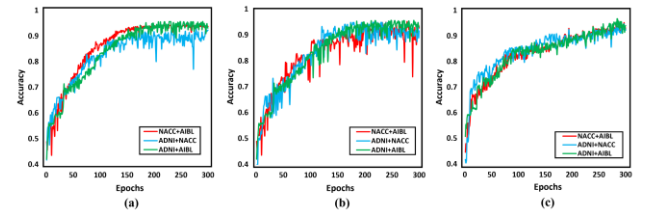


Fig. 2 Training accuracy for the proposed DCL on source domains (a) AD vs NC (b) AD vs MCI (c) MCI vs NC

Our method not only achieved the highest accuracy of 94.53% but also demonstrated outstanding sensitivity and specificity, recording values of 94.82% and 94.85% respectively. This further demonstrates the method's

Table 2. Performance evaluation (%) of the proposed method and chosen baseline methods on the target datasets.

Models	Source Dataset	Target Dataset	AD vs NC			AD vs MCI			MCI vs NC		
			Acc.	Sen.	Spe.	Acc.	Sen.	Spe.	Acc.	Sen.	Spe.
[15]	ADNI + AIBL	NACC	90.47	91.74	90.96	89.68	90.82	91.11	90.21	90.31	91.21
[4]			89.55	91.83	91.49	88.84	89.15	89.97	87.88	88.94	89.12
[11]			89.56	88.74	89.96	86.83	87.35	88.17	86.51	86.28	87.25
[16]			93.15	93.53	93.49	91.82	90.12	90.48	89.13	89.09	89.89
Ours			94.28	94.47	94.85	92.97	92.01	92.87	90.34	90.43	90.75
[15]	ADNI + NACC	AIBL	89.46	91.46	89.08	88.12	88.23	88.04	88.92	88.87	89.03
[4]			87.71	87.38	87.13	86.13	86.87	86.94	86.75	86.63	86.86
[11]			88.91	88.54	90.75	91.18	90.04	87.54	87.89	87.14	87.93
[16]			90.94	90.77	90.13	89.94	89.57	88.83	88.12	88.25	88.46
Ours			92.27	92.89	92.99	91.41	90.87	91.02	90.18	90.27	90.38
[15]	NACC + AIBL	ADNI	90.89	90.23	90.45	89.16	89.13	89.24	88.29	87.72	88.81
[4]			89.12	91.87	91.75	89.92	87.98	90.16	87.89	86.87	87.95
[11]			89.47	89.17	89.78	87.85	87.48	88.76	86.99	86.84	86.47
[16]			93.18	93.03	93.47	91.03	90.17	90.97	89.89	89.68	89.72
Ours			94.53	94.82	93.89	92.11	91.94	91.85	90.87	90.64	90.89

consistent and robust performance across different sites, emphasizing its generalization ability.

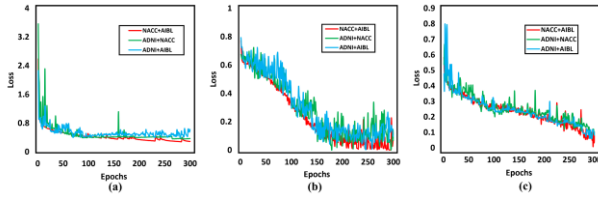


Fig.2. Training loss for the proposed DCL on source domains (a) AD vs NC (b) AD vs MCI (c) MCI vs NC

4. Discussion and Limitation

In this paper, we proposed a novel approach to multi-site AD classification, leveraging contrastive learning in the latent space for enhanced domain generalization across multiple sites. The proposed method, called Domain Contrastive Learning (DCL) facilitates the automatic diagnosis of AD and its prodromal stages. DCL significantly outperformed existing methods, as detailed in Table 2. This advantage is primarily attributed to its unique ability to extract domain-invariant features, greatly enhancing its generalization capabilities. This was particularly evident in the AD vs. NC classification task, where DCL achieved remarkable accuracy, sensitivity, and specificity. The high performance in AD vs. NC classification might stem from the relative ease of distinguishing AD from NC compared to classifying MCI stages. MCI classification is challenging

due to the variability in patient conditions, including stable and progressive stages. This complexity, a common issue in the field [4], suggests a need for separate consideration of MCI stages in future models. The capability to learn domain-invariant features positions DCL as a robust method for AD diagnosis in diverse clinical settings, addressing a crucial need in multi-site Alzheimer's classification. However, the study is not without its limitations. A notable challenge is the class imbalance in the datasets used. Future research will focus on incorporating a broader range of datasets and employing advanced augmentation techniques to further validate and refine the model's performance.

Moreover, integrating additional modalities like MRI and PET scans could potentially improve diagnostic accuracy.

5. Conclusion

In this paper, we presented a Domain Contrastive learning (DCL) framework to tackle the domain generalization problem in Multi-site Alzheimer's classification. We introduced a self-supervised contrastive learning strategy in the latent space of our autoencoder to learn domain-invariant representations of the data belonging to the three main classes: AD, NC and MCI irrespective of the domains/sites the images were collected. These representations are then sent to a classifier to achieve generalizability. The rigorous evaluation of our model on brain images from three distinct imaging sites, employing a Domain Generalization training protocol, has demonstrated

DCL's remarkable robustness and generalizability. Experimental results on the three MR image datasets outperformed the compared state-of-the-art generalization methods for AD classification. In the future, it is of interest to improve our DCL framework to accommodate more datasets as well as handle data imbalance issues that affect multisite multiclass MR image classification.

Acknowledgements

This research was supported by the Science and Technology Project of Sichuan (Grant NOs.2022ZHCG0033, 2021YFG0314, 2020YFG0459), and the National Natural Science Foundation of China (Grant NOs. U19A2078).

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